Project 2: Video Classification

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1. Introduction

Recognising an action from video means combining what each frame shows with how it changes over time. We study this on a ten-class workout subset of UCF101 [7], where classes such as *PushUps* and *WallPushups*, or *JumpingJack* and *JumpRope*, share poses and scenes but differ in motion. We ask three questions. (i) Where should a CNN combine the frames: after classifying each one, after pooling their features, at the input, or throughout a 3D network (Task 4.1)? (ii) How much do the original splits, in which the same people appear in training and evaluation, inflate these accuracies (Task 4.2.1)? (iii) Does an optical-flow stream help once identity cues are removed (Task 4.2.2)? So that the comparison is about the fusion point rather than the network, all models share one backbone and one training recipe.

Related work. Karpathy *et al.* [3] compared single-frame, early, late, and slow fusion of 2D CNNs and found single-frame models surprisingly strong. C3D [9] learns spatiotemporal features with 3×3×3 kernels, and I3D [1] initialises such kernels by inflating trained 2D filters. Two-stream networks [5] add a temporal stream on stacked optical flow and average the two streams' softmax scores; TSN [10] initialises the flow network from RGB weights. We apply these ideas at small scale, training on only 500 videos.

2. Data and setup

Splits. The original splits contain 500/120/120 training/validation/test videos, balanced over ten classes. They were taken from Kaggle and place clips of the same person in several splits. The corrected splits, which keep each person in a single split, contain 500/120/120 videos. Each video is represented by ten uniformly sampled RGB frames. The corrected data also provides nine RAFT [8] flow fields between consecutive frames, stored as colour-wheel images (hue encodes direction, saturation encodes magnitude).

Preprocessing and augmentation. Frames are resized to 112×112 and standardised with ImageNet statistics; flow images are inverted so that zero motion (white) maps to 0. During training, each clip receives one random resized crop (scale 0.5–1), horizontal flip, and colour jitter, shared by all its frames so that augmentation does not create spurious motion. Flips are disabled for flow, because mirroring a colour-coded field would not reverse the encoded direction.

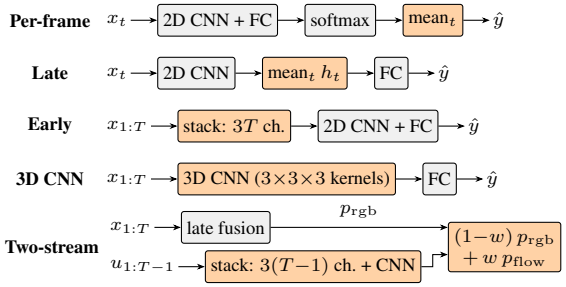


Figure 1. Where each model combines the T=10 frames (orange). All models share the same 2D backbone (x_t: RGB frame, h_t: its feature vector, u_t: colour-coded RAFT flow from frame t to t+1). The two streams are trained separately and fused by weighting their softmax scores.

Training and evaluation. All models use AdamW [4] (learning rate 10^{-3}, weight decay 0.05), three warm-up epochs, and cosine decay over a fixed 60 epochs. Every update sees 80 frames, either eight videos or, for the per-frame model, 80 independent frames, so all models receive the same number of updates and frames. We keep the checkpoint with the highest validation video accuracy and evaluate each selected model once on the test set. One validation video corresponds to 0.83 percentage points (pp). All results use a single seed (42).

3. Models

Shared backbone. Every variant uses the same VGG-style [6] network: four stages, each of two 3×3 convolution–batch-norm [2]–ReLU layers (64/128/256/256 channels) followed by 2×2 max pooling. Global average pooling gives a frame feature $h(x_t) \in \mathbb{R}^{256}$, and dropout (0.5) precedes a linear classifier (2.33 M parameters).

Why this backbone. To measure fusion rather than optimisation, every variant must be able to fit its training set. Our first backbone, three convolutions without normalisation, failed this check. Per-frame aggregation reached 90.8% validation accuracy, but early fusion and the 3D CNN fitted only 29% and 42% of their training videos. Their low validation accuracy (22.5% and 32.5%) therefore reflected underfitting, not missing temporal information. The current backbone adds batch normalisation, which stabilises optimisation, and a fourth stage, which enlarges the receptive field from 22 px (10% of the frame) to 76 px (68%). Together with the warm starts below, these changes let every variant fit its training set (Sec. 4).

Fusion variants (Fig. 1). With logits $z_t = Wh(x_t) + b$ for frame t and $T = 10$:

$$p_{\text{frame}}(y \mid V) = \tfrac{1}{T} \textstyle\sum_t \operatorname{softmax}(z_t)_y, \quad (1)$$

$$p_{\text{late}}(y \mid V) = \operatorname{softmax}\left(W \tfrac{1}{T} \textstyle\sum_t h(x_t) + b\right)_y. \quad (2)$$

The per-frame model trains on single frames with their video's label; late fusion trains on whole videos. Both ignore frame order. Early fusion stacks the frames into a 30-channel image, so only its first convolution is wider (2.35 M parameters) and it sees frame order from the first layer. The 3D CNN uses the same layer layout with $3\times3\times3$ kernels; pooling keeps time in the first two stages and halves it in the last two (T: $10 \to 10 \to 5 \to 2$), so its final features span the whole clip (6.98 M parameters).

Warm starts. Training video models from scratch on 500 videos is difficult, so late fusion, early fusion, and the 3D CNN start from the trained per-frame model of the same split. Late fusion copies its weights. Early fusion repeats the RGB input kernel over the T frames and divides it by T [10]. The 3D CNN inflates every kernel, $W^{3D}_{:,:,\tau} = W^{2D}/3$ [1]. For a static clip, both inflated models initially reproduce the per-frame features (approximately, for the 3D CNN), so any gain must come from motion learned during training.

Two streams. The temporal stream is an early-fusion network over the nine stacked flow fields (27 input channels) [5]. The RGB stream is the late-fusion model trained on the corrected splits. As in the original two-stream work, the streams are trained separately and combined as $p = (1 - w)\, p_{\text{rgb}} + w\, p_{\text{flow}}$, with $w \in \{0, 0.05, \ldots, 1\}$ chosen on validation; we also report $w = 0.5$. As an alternative, a jointly trained dual-stream model concatenates both streams' features before one linear layer (4.67 M parameters), with its RGB encoder initialised from the per-frame model.

Table 1. Video accuracy (%) of the four fusion variants on the original (leaky) and corrected splits. Δ: test-accuracy change in pp when identity leakage is removed. Video models are warm-started from the per-frame model of the same split.

		Original		Corrected		
Model	Params	Val	Test	Val	Test	Δ
Per-frame	2.33 M	93.3	91.7	40.8	35.0	−56.7
Late fusion	2.33 M	93.3	93.3	43.3	38.3	−55.0
Early fusion	2.35 M	89.2	91.7	40.0	36.7	−55.0
3D CNN	6.98 M	92.5	90.8	34.2	41.7	−49.2

Table 2. Two-stream results on the corrected splits (video accuracy, %). The fusion weight w is selected on validation only; equal weighting ($w = 0.5$) gives the same accuracies.

Model	Val	Test
RGB stream (late fusion)	43.3	38.3
Flow stream (stacked flow)	55.8	49.2
Score fusion, $w = 0.55$ (val.-selected)	58.3	54.2
Joint dual stream	55.0	54.2

4. Experiments

4.1. Where to fuse time (Task 4.1)

On the original splits all four variants reach 89–93% validation and 91–93% test accuracy (Tab. 1). The largest test gap, late fusion against the 3D CNN, is three videos (2.5 pp), which is about one binomial standard error at this accuracy. No fusion point is therefore clearly better, matching Karpathy *et al.*'s observation that single-frame models are hard to beat [3]. Early fusion and the 3D CNN now fit the training set (98% final training accuracy, Fig. 2, left), passing the check that the first backbone failed.

4.2. Information leakage (Task 4.2.1)

Because the original splits share people and settings across splits, a model can score well by recognising them instead of the action. Retrained with the same recipe on the corrected splits, every RGB model loses 49–57 pp of test accuracy, falling to 35–42%, while its training accuracy still reaches ≥99% and its validation loss stays near 3 (Fig. 2, right). The models had memorised people and settings. Their ranking is not stable: the 3D CNN is the weakest on validation (34.2%) but the strongest on test (41.7%), where it drops the least. With a standard error of about 4.5 pp at 40% accuracy, the four RGB variants remain indistinguishable on the corrected splits.

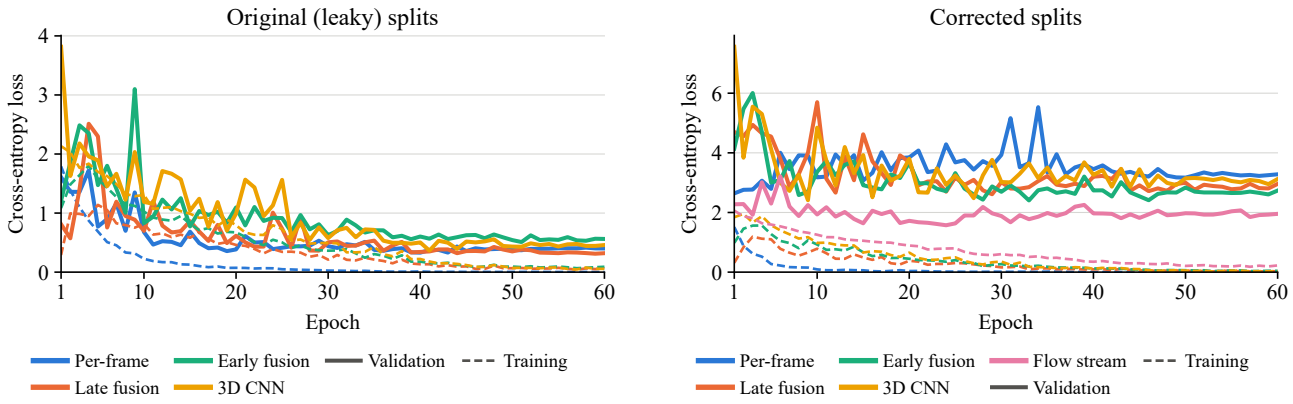


Figure 2. Training (dashed) and validation (solid) cross-entropy loss per epoch on the original (left) and corrected (right) splits, seed 42. Per-frame training loss is computed on single frames; all validation losses are video-level.

4.3. Optical flow (Task 4.2.2)

The flow stream alone reaches 55.8% validation and 49.2% test accuracy, 12.5 and 10.8 pp above the RGB stream, and it fits the training set more slowly (83% at its selected epoch). Motion therefore transfers across people far better than appearance. Score fusion adds a further 2.5 pp on validation and 5.0 pp on test (54.2%), because the streams fail on different classes. On test, flow recognises JumpingJack (11/12 videos, against 5/12 for RGB), BodyWeightSquats (8 against 1), and PushUps (8 against 4), whereas RGB identifies TrampolineJumping from its scene (10 against 2). Fusion only helps once $w \geq 0.3$: below that, the overfitted, overconfident RGB stream (validation NLL 2.91, against 1.86 for flow) dominates the average. Joint training matches score fusion on test (54.2%) but is 3.3 pp lower on validation (55.0%), so neither way of combining the streams is clearly better.

5. Discussion

What the comparison isolates. Because every variant shares the backbone, recipe, update budget, and initial per-frame weights, differences in Tab. 1 reflect where frames are combined, not which network a model received. Under this control, where an RGB network fuses time barely matters: the variants are within noise on the leaky splits, and on the corrected splits they stay within 7 pp of each other on test (9 pp on validation), with the two splits disagreeing about their order. From 500 videos, the RGB networks did not learn motion features that reliably generalise to new people.

Appearance shortcuts. A drop of 49–57 pp for every RGB model, with near-perfect training accuracy, means that on the leaky splits the task largely reduces to recognising people and settings. Leaky evaluation therefore overstates accuracy by more than a factor of two, and a model chosen on leaky splits says little about how it will perform on new people.

Motion. Optical flow removes most appearance cues by construction, which is why the flow stream overfits more slowly and generalises best. RGB still adds complementary evidence: the streams fail on different classes, so combining them helps whether the streams are fused at the score level or trained jointly.

Limitations. Results use one seed and 120 videos per evaluation split; one video is 0.83 pp, and the binomial standard error near 50% accuracy is about 4.6 pp. Validation accuracies are also optimistic, because the checkpoint is the best of 60 noisy epochs: on the corrected splits, the RGB models average only 28–37% over their last ten epochs. Colour-coded flow is a lossy encoding of the displacement vectors and rules out flip augmentation.

6. Conclusion

We compared four ways of fusing time in a shared 2D/3D backbone, measured the effect of person leakage, and added an optical-flow stream. On the leaky splits all fusion points perform equally well (91–93% test accuracy). Person-disjoint splits reveal that this performance came mostly from recognising people: every RGB model loses 49–57 pp of test accuracy, and the variants become indistinguishable at 35–42%. Motion is the cue that transfers: a flow stream on nine stacked RAFT fields reaches 49.2% test accuracy, 7.5 pp above the best RGB model, and fusing it with the RGB stream reaches 54.2%, 15.8 pp above RGB alone. The main limitation is overfitting on 500 training videos; pre-training the RGB stream and averaging over seeds are the natural next steps.

References

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Statement on use of generative AI

OpenAI Codex assisted with project scaffolding, configuration files, data-loading and the first model implementations and automated checks. Anthropic's Claude (Claude Code) reviewed the model designs, implemented augmentation, warm-start inflation, score fusion and evaluation code.